Coastal Monitoring Station Specifications

This document provides technical specifications for the twelve coastal monitoring stations that comprise the Eastern Seaboard SST Monitoring Network. Each station consists of a moored buoy equipped with temperature sensors, conductivity probes, and meteorological instruments. The network has been operational since 1985 and provides continuous high-frequency observations of sea surface conditions along the eastern coast.

Station Deployment Protocol

Monitoring stations are deployed at fixed locations selected for their representativeness of regional oceanographic conditions. Station positions are chosen to avoid local influences such as river discharge, industrial cooling water outflows, and shallow reef effects that might bias temperature readings relative to the broader coastal water mass. Each station is positioned in water depths between fifteen and thirty meters, far enough offshore to represent open coastal conditions while remaining accessible for routine maintenance operations conducted monthly during summer and quarterly during winter.

Instrumentation

The primary temperature sensor at each station is a platinum resistance thermometer with an accuracy of plus or minus zero point zero one degrees Celsius and a response time of less than five seconds. Sensors are mounted at one meter depth on the mooring line, providing a measurement representative of the surface mixed layer under most conditions. Backup sensors at three meters and five meters depth provide redundancy and allow detection of near-surface stratification events during periods of strong heating.

Conductivity sensors measure salinity with an accuracy of plus or minus zero point zero zero five practical salinity units. Combined with temperature measurements, conductivity data allow calculation of seawater density, which is used to identify water mass boundaries and mixing events. All sensors undergo laboratory calibration before deployment and post-recovery calibration checks to quantify any drift during the deployment period.

Data Transmission and Quality Control

Station data are transmitted hourly via satellite telemetry to the central data processing facility. Real-time quality control algorithms flag obvious errors such as sensor failures, biofouling artifacts, and transmission corruption. These flagged data points are excluded from real-time products but retained in the archive for potential recovery during delayed mode processing. Delayed-mode quality control, conducted within six months of data collection, applies more sophisticated algorithms including comparison with satellite derived sea surface temperature, neighboring station cross-checks, and statistical tests for consistency with climatological expectations. The final quality-controlled dataset receives a quality flag for each observation indicating whether it passed all tests, failed specific tests, or was interpolated to fill gaps created by instrument downtime.

Data gaps resulting from instrument failures or maintenance periods are filled using optimal interpolation methods that account for spatial correlation with neighboring stations and temporal autocorrelation within the station record. Interpolated values are clearly flagged in the dataset and excluded from trend analyses to avoid introducing artificial signals into long-term monitoring records.